

3.3 Objectives

Objective is to undertake an environmental assessment of the proposed subproject and prepare Environmental Management Plan, with cost estimates for implementation, institutional responsibility for implementation, monitoring and reporting of Environmental Management Plan.

4. Project Description

The existing drainage network in Greater Chennai Corporation has many missing links due to which flooding is occurring in certain areas. As such based-on ground Condition and initial assessment by officials of GCC a list of Missing links SWD were identified. Hence to ensure an Integrated SWD network, there is a need to provide the drains for the missing links connecting the Upstream & Downstream drains & to serve unserved areas.

4.1 Study Area

The study area covers 6 Zones of the Greater Chennai Corporation, Zones covered in the study area are, Zone 4 – Tondiarpet, Zone 6 - Thiru Vi Ka Nagar, Zone 8 - Anna Nagar, Zone 9 - Teynampet, Zone 10 – Kodambakkam, Zone 13 – Adyar. The list of various missing link drains proposed are detailed below:-

S. No	Zn No.	Dn No	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
1	IV	38	Nethaji Nagar Market Street	900x900	205	D&R	Nethaji Nagar Main Road
2	IV	38	Nethaji Nagar Main Road	900x900	240	D&R	Vinoba Nagar
3	IV	36	Thiruvalluvar Nagar 4th Main Road	900x900	55	D&R	Thiruvalluvar Nagar 5th Main Road
4	IV	36	Thiruvalluvar Nagar 1st cross street.	900x900	110	D&R	TVK 2 nd Link Road
5	IV	36	MKB Nagar 18th west cross st	900x900	288	Link	West Avenue service Road
6	IV	37	MKB Nagar 2nd main road west dn 36	900x900	230	D&R	18 th west cross st MKB Nagar
7	IV	44	Thiruvalluvar street	900x900	320	D&R	Jawahar Canal

S. No	Zn No.	Dn No	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
8	IV	44	Thiruvalluvar Lane	900x900	70	Link	Kattabomman Street
9	IV	44	Rajangam Street	900x900	100	Link	Maa Poo si 1 st street
10	IV	44	Mathiyalagan street	900x900	170	Link	Rajangam Street
11	IV	44	NSK street	900x900	125	Link	Rajangam Street
12	IV	45	Sundaram 3rd street	900x900	210	D&R	Sundaram Main Road
13	IV	45	Giramiyar street	900x900	170	D&R	
14	IV	45	B B Road 1st street	900x900	60	D&R	B B Road
15	IV	45	B B Road 2nd street	900x900	80	D&R	B B Road
16	IV	45	B B Road 3rd street	900x900	165	D&R	B B Road
17	IV	45	Balamurugan street	900x900	320	D&R	Melpatti ponnappan Street
18	IV	45	Sriram nagar	900x900	125	D&R	Melpatti ponnappan Street
19	IV	45	Ganesapuram Main Road	900x900	215	D&R	Dr Ambedkar Collage Road
20	IV	46	Palla 1 to 4 streets	900x900	1200	D&R	Muthu Street
21	IV	46	Moorthingar Road (from paravai park to CMS moorthingar road) at western side	900x900	200	Link	Chennai Middle School
22	IV	47	K.H.Road	900x900	350	D&R	Sathyamurthy St
23	IV	47	Sathyamurthy street	900x900	225	D&R	Motchaporan 2 nd Street
24	IV	47	Motchapuram 2nd St	900x900	180	Link	B Canal
25	IV	47	Meenambal Nagar 4th st	900x900	150	Link	Meenambal Nagar Cross st
26	IV	47	Meenambal Nagar 5th st	900x900	155	Link	Meenambal Nagar Cross st
27	IV	47	Thyagappast	900x900	250	D&R	Kannan Street
28	IV	47	Ekambaram St	900x900	255	D&R	Kannan Street

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
29	IV	47	Kannan St	900x900	450	D&R	Jewanmul sowcar street
30	IV	47	Jawanmulsowcarst	1200x1200	270	D&R	Kannan Street
31	IV	47	Meenambal Nagar Cross st	900x900	120	D&R	Meenambal Nagar Cross 6 th st
32	V	49	Kappal Polu Street (from Jn of Sollaiappan Street to BalaArunachalam street)	600x750	410	D&R	Bala Arunachalam street
33	V	50	Mannarsamy Koil Street (From East Kalmandapam Road to Andiyappan Street)	900x900	600	D&R	Andiyappan Street
34	V	51	N.N. Garden 8th Lane (N.N.Gardern 8th Lane to Narassiar Street)	600x750	300	D&R	Narassiar Street
35	V	53	Kathbada Main Road (From TH Road Service Road to existing drain at kathbada Road)	900x900	600	D&R	Kathbada Road
36	V	58	Vepery High Road (North Side) (from Madox street to Sydenahams road)	900x900	750	D&R	Sydenahams road
37	V	58	Jothi Venkatchalam Street (from vepery high road to EVR Salai)	900x900	600	D&R	EVR Salai
38	V	58	Valliammal Road (from Alagappa Road to Ritherdon road)	900x900	300	D&R	Ritherdon road
39	V	60	Mooker Nallumuthu street (from Mooker Nallumuthu street to Prakasam salai)	600x750	450	D&R	Prakasam salai

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
40	V	60	Jaffer Sarang Street (from Jaffer Sarang Street to Mannady street)	600x750	290	D&R	Mannady street
41	V	60	RajajiSalai (from Clive Battery to NRT Bridge)	1200x1200	400	D&R	NRT Brid
42	V	61	Kennet lane (from Gandhi Irwin road to Patheon road)	600x750	300	D&R	Patheon road
43	V	61	Montieth Road	900x900	250	D&R	
44	V	62	SingannaChetty Street (from Guruvappa Street to Wallers street)	900x900	350	D&R	Wallers street
45	V	63	Begam Sahib 2nd, 3rd & 4th Street	600x750	400	D&R	GP Road
46	V	63	Burrah Sahib Street	600x750	300	D&R	Chellapillayar Street
47	VI	78	Asthabujam street	900x900	475	Link	Choolai High Road
48	VI	78	Angallamankoil street	1200x1200	290	D&R and Link	Choolai High Road
49	VI	72	Brightons Road	900x900	280	Link	Otteri Nallah
50	VI	76	Venkatesabaktha street	900x900	260	Link	Perambur Barracks Road
51	VI	76	Sachidanandam street	900x900	160	Link	Perambur Barracks Road
52	VI	70	Govindan street	900x900	265	Link	Cooks Road
53	VI	71	Jamaliya Link road	900x900	170	D&R and Link	Perambur High Road South
54	VI	70	Perambur High road South	1500x1500	880	D&R and Link	Cooks Road

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
55	VI	72	Basin powerhouse Road	900x900	450	Link	Pulianthope high Road
56	VI	70	Madhavaram high road	900x900	375	D&R and Link	Eganajipuram High "Road
57	VI	65	Senthilnagar 13 th street	900x900	140	Link	Paper Mills Road
58	VI	69	Bunder garden 1 st street	900x900	170	D&R and Link	Madhavaram High Road
59	VI	69	Bunder garden 2nd street	900x900	210	Link	Madhavaram High Road
60	VI	69	Neelam garden	900x900	230	D&R and Link	Madhavaram High Road
61	VI	68	Maduraisamy madam	1200x 1200	510	D&R and Link	Pallavan Salai
62	VI	65	Srinivasanagar 1st main road	1200x 1200	30	D&R and	Paper Mills Road
63	VI	65	Venus nagar 1st main road	900x900	10	D&R and Link	Paper Mills Road
64	VI	66	Karthick street	900x900	20	D&R and Link	Paper Mills Road
65	VI	66	Sakthivinayagarkoil street	900x900	120	D&R and Link	Paper Mills Road
66	VI	66	Lakshmannagar east street	900x900	60	D&R and Link	Paper Mills Road
67	VI	66	Jawaharnagar 6th main road	1200x 1200	12	D&R and Link	Paper Mills Road

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
68	VI	68	State Bank colony road	900x900	370	D&R and Link	Madhavaram High Road
69	VI	68	Paper mills road	1200x 1200	287	D&R and Link	Eganajipurin Gigh "Road
70	VIII	94	South Jagannathan Nagar 1st Main Road	900x900	660	D&R and Link	Redhills Villivakkam Road
71	VIII	94	South Jagannathan Nagar 2nd cross Street	900x900	175	D&R and Link	Redhills Villivakkam Road
72	VIII	94	Periyar Nagar Main Road	900x900	310	D&R and Link	Redhills Villivakkam Road
73	VIII	94	South Jagannathan Nagar 3rd Main Road	900x900	400	D&R and Link	Redhills Villivakkam Road
74	VIII	95	Mannadi Othavadai Street	900x900	300	D&R and Link	Seeyalam Canal
75	VIII	95	Agathiar Nagar 31st Street	900x900	275	D&R and Link	Agathiyar nagar 7 th street to TVS canal
76	VIII	95	Kumaraswamynagar 1st main road	1200x 1200	310	D&R and Link	Kumarasamy nagar
77	VIII	95	Thiruvengkadiah 1st Street	900x900	125	D&R and Link	MTH Road
78	VIII	95	North Thirumalai Nagar 2nd Street	900x900	190	D&R and Link	Dr. Ambedkar Road (ICF)

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
79	VIII	95	South ThirumalaiNagar 1st Street	900x900	120	D&R and Link	Dr. Ambedkar Road (ICF)
80	VIII	95	South ThirumalaiNagar 3rd Street	900x900	120	D&R and Link	Dr. Ambedkar Road (ICF)
81	VIII	95	South ThirumalaiNagar Main Street	900x900	200	D&R and Link	Dr. Ambedkar Road (ICF)
82	VIII	101	C Block 6th,7th and 8th street.	900x900	265	D&R and Link	Anna Nagar 2 nd main road
83	VIII	105	Perumalkoil garden	900x900	80	Link	Perumal Koil Street
84	VIII	105	Kalkinagar	900x900	200	D&R and Link	SA Staff colony 7 th street
85	VIII	105	Duraipillai street	900x900	150	D&R and Link	Mariamman Koil Street
86	VIII	105	Ashoka street	900x900	268	D&R and Link	Water tank road near A Block
87	VIII	105	MMDA colony Main road	900x900	400	D&R and Link	Thirukuralam Street
88	VIII	108	Padmanaba Nagar Main Road	900x900	260	D&R and Link	Tamilasu Veedi st
89	VIII	108	Padmanaba Nagar 2nd main Road	900x900	260	D&R and Link	Tamilasu Veedi st
90	VIII	105	Jaganathan 1st main road	900x900	130	Link	Flowers Road

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
91	VIII	105	Vinayagapuram 2nd street.	900x900	165	Link	100feet road
92	VIII	94	Sidconagar 2nd main road	1200x 1200	400	D&R and Link	Kumarasa Canal
93	IX	109	Khan Street	600x750	236	Link	Choolaimedu High Road
94	IX	110	Kodambakkam High Road Service Road North	900x900	75	D&R	Mahalinga-puram Main Road
95	IX	111	Pattulas Road	900x900	10	D&R	Anna Salai
96	IX	111	Club House Road	900x900	15	D&R	Anna Salai
97	IX	112	New Vellalar street	900x900	160	Link	Vellalar Street
98	IX	112	Vellalar street	900x900	240	Link	N.S.K. Salai
99	IX	112	Sivan koil cross street	900x900	140	Link	Sivan Koil Street
100	IX	113	Thirumallaipillai road	900x900	150	Link	Habibulla Road
101	IX	113	Shenoy road	900x900	300	D&R	Tank Bund Road
102	IX	113	Dharmapuram main Road	900x900	300	D&R	Bala muthu krishnan Street
103	IX	115	Devaraj street	900x900	365	D&R	Triplecane High Road
104	IX	117	South boag road	1200x 1200	1162	D&R	Habibulla Road
105	IX	118	Cathedral Road	900x900	900	D&R	TTK Road
106	IX	119	V.M.Street	900x900	600	D&R	Avvai shanmugam Salai
107	IX	120	Dr.Natesan Road	900x900	300	D&R	Besant Road
108	IX	121	Dr.Radhakrishnanansalai	900x900	520	Link	Dr. Radhakrishnan Salai

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
109	IX	122	Cenataph road	900x900	870	D&R	Charmers Road
110	IX	122	Turnbulls road	900x900	100	Link	Habibulla Road
111	IX	123	Rama Rao Road	600x750	200	Link	Habibulla Road
112	IX	124	R.K.Mutt Road	900x900	250	Link	Santhome High Road
113	IX	125	KamarajarSalai	1200x 1200	250	Link	Kaneeswar Pogoda Street
114	IX	126	Devanathan Street	900x900	300	D&R	St.Marys Road
115	X	127	Vinayagam street	900x900	200	D&R	Kamarajar Salai
116	X	128	West Vanniar Street	1200x 1200	220	D&R	Thiruvalluvar 2 nd Street
117	X	128	Srinivasa Street	1200x 1200	180	D&R	West Vanniar Street
118	X	129	Senthilandavar street	900x900	160	D&R	K K Salai
119	X	129	Arunachalam road (western side)	1200x 1200	1000	Link	Kamarajar Street Gandhi Nagar
120	X	130	West Sivan Koil Street	900x900	110	D&R	Arcot Road
121	X	130	Gangaiammankoil street (saligramam)	900x900	190	D&R	Nerkundram Pathai
122	X	130	South perumalkoil street	900x900	200	D&R	Pillayar Kovil Street
123	X	138	Anna Main Road (South side)	1200x 1200	950	D&R	MGR Canal Connectivity drain
124	X	138	Ashok pillar road (southern side)	1200x 1200	400	Link	100 Feet Road
125	X	132	Saidapet road	900x900	300	D&R	100n feet Road

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
126	X	133	V.O.C.Nagar 1st Main Road	900x900	300	D&R	Arcot Road
127	X	133	Palayakkaran Street	900x900	300	D&R	Arcot Road
128	X	142	West Jones Road	900x900	500	D&R	Kanniyamman Koil Street
129	XIII	172	Lake View Road From Clri Compound To South 'B' Canal	1500x1500	500	Link	South B Canal
130	XIII	173	Crescent Avenue, East Kesavaperumal	900x900	600	D&R	Central B Canal
131	XIII	173	DGS salai (From Petrol bunk to Janagi MGR college)	900x900	150	Link	Adyar River
132	XIII	174	Velachery Main Road (Near Halda junction)	900x900	515	Link	Adyar River
133	XIII	174	Velachery Main Road (From Castle apartment to Bye pass signal)	900x900	400	D&R	Velachery Lake
134	XIII	175	Thiruvankadam street, K.B.Nagar 3rd main road	900x900	600	D&R	South B Canal
135	XIII	175	Govindarajapuram 1st street, 2nd street	600x750	335	Link	South B Canal
136	XIII	175	K.B nagar 3rd cross street	1200x1200	1200	D&R	South B Canal
137	XIII	175	Gandhi nagar 1st Crescent park road	900x900	300	D&R	Adyar River
138	XIII	177	MGR Nagar 9th Street	600x750	320.00	Link	Veerangal Odai
139	XIII	178	Bhel Sakthi Nagar main road	600x750	170	Link	Verangal Odai
140	XIII	178	Orandiamman Koil main road	600x750	450	Link	Velachery Lake

S. No.	Zn No.	Dn. No.	Name of the Road	SWD size required (mm)	Length (m)	Link or D&R	Disposal point of the purpose same
141	XIII	179	Tansi Nagar 15th Street, 16th Street	900x900	270	Link	Pallikaranai Marsh land
142	XIII	179	Natarajan Street, Gandhi Road, Thiruveethi amman koil st, VGP avenue, Seethapathy Nagar 2nd cross street.	900x900	1500	Link	South B Canal
143	XIII	179	LIC colony 1st street	600x750	70	Link	South B Canal
144	XIII	179	Seshadripuram 3rd cross street	600x750	150	Link	Pallikarunai Marsh land
			Total		44883 m		

Table 7 Various missing link drains proposed

Most of the Missing link drains are proposed along the roads. Reconstruction/Augmentation of existing drains, which were found inadequate, has been proposed in this project. RCC rectangular cast-in-situ drains have been proposed for construction.

Other components are construction of cross drainage works, Rainwater Harvesting structures, Silt catch traps, preventing dumping of solid wastes in the canals to prevent entry of solid wastes into the system, etc.

The Detailed Project Report has been prepared for providing missing links Storm Water Drain in the core city of Greater Chennai Corporation which covers the Zones Viz. Zone 4 (Tondiarpet), Zone 5 (Royapuram), Zone 6 (Thiru. Vi. Ka Nagar), Zone 8 (Anna Nagar), Zone 9 (Teynampet), Zone 10 (Kodambakkam) and Zone 13 (Adyar) by undertaking a technical, financial, economic, environmental and social studies of the project area.

The proposals include the demolition and reconstruction of the existing storm water drain system in dilapidated conditions and the damaged drains constructed decades ago with brick and mortar and also providing missing link drains in the missing links portion so as to establish a integrated storm water drain network wherever the link is missing it is proposed to establish an effective and integrated drainage system with